

TANU CHAUDHARY

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PROFESSIONAL SUMMARY

Software Engineer with strong foundations in DSA, OOP, and system design, experienced in building scalable backend services and full-stack web applications using Java, Python, Spring Boot, and the MERN stack. Proven ability to design RESTful APIs, optimize concurrent systems, and deliver measurable engineering impact in agile development environments.

EDUCATION

Vellore Institute of Technology, Chennai

Expected 2027

Master of Computer Applications | CGPA: 8.62

Galgotias University, Noida

2022 – 2025

Bachelor of Computer Applications IO AI/ML | CGPA: 8.6

EXPERIENCE

Google Generative AI Virtual Intern — AICTE

Jul 2024 – Sep 2024

- Designed AI automation pipelines using prompt engineering, reducing manual query handling by 30%.
- Built foundational understanding of LLM inference pipelines and production-grade AI system architecture through hands-on project work.

PROJECTS

Smart City Civic Management Platform | MERN Stack, REST APIs, Node.js, MongoDB | [Github](#)

- Architected a full-stack platform with React.js frontend and Node.js/Express.js backend, supporting real-time complaint tracking across multiple civic service modules.
- Designed 15+ RESTful API endpoints with optimized query patterns on MongoDB, reducing average response latency by 35% and supporting 500+ concurrent users.

Crime Prediction & Analytics Engine | Python, XGBoost, Random Forest, FastAPI | [Github](#)

- Engineered an ML inference pipeline over 50,000+ crime records, achieving 89% F1-score for hotspot prediction using ensemble models (XGBoost, Random Forest, Decision Tree).
- Integrated models with a FastAPI backend service; optimized data preprocessing and batch-inference strategies to reduce prediction processing time by 25%.

AI Desktop Assistant | Python, OpenCV, Gemini API, Speech Recognition, REST APIs | [Github](#)

- Built a multi-modal desktop agent with face authentication (OpenCV), LLM-backed responses (Gemini API), and TTS automating 20+ OS-level tasks including app control and web interaction.
- Designed event-driven architecture with concurrent task threads, ensuring <200ms response latency for real-time voice and vision processing.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, PHP

Backend & Distributed Systems: Spring Boot, FastAPI, Node.js, Express.js, REST APIs, Microservices, Concurrency, System Design, Agile

Frontend: React.js, Angular, HTML, CSS, AJAX, jQuery

Databases: MySQL, MongoDB

CS Fundamentals: DSA, OOP, DBMS, OS, Computer Networks

Machine Learning & AI: Machine Learning, Deep Learning, Data Preprocessing, Model Evaluation, Transformers

Tools: Git, Docker, AWS (EC2, S3), Postman

CERTIFICATIONS

Web Development Fundamentals — IBM | [Link](#)

Database Programming with SQL — Oracle | [Link](#)

AI & ML Engineering — Microsoft | [Link](#)

Machine Learning Foundations — AWS | [Link](#)

ACHIEVEMENTS

- Research: Quantum-Inspired Multi-Agent Learning for Blue Carbon Modeling — Under Review, Springer.
- Smart India Hackathon Pre-Qualifier — 2023 & 2024.